

MONTREAL MIRABEL INTL, QC, Canada

WMO: 716278

Lat: 45.6685N

Lon: 74.0322W

Elev: 270

StdP: 14.55

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.2	-9.0	-25.2	1.4	-12.6	-20.1	1.8	-7.8	21.5	22.9	19.6	20.2	5.5	240	0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	85.5	71.7	82.6	69.7	80.0	68.1	73.8	82.0	71.9	79.5	70.1	77.2	8.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.2	116.2	78.7	69.3	108.9	76.4	67.6	102.6	74.6	37.6	82.0	35.9	79.6	34.3	77.0	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min		Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	15.7	13.7		-20.6	89.9	6.0	2.7	-24.9	91.9	-28.4	93.5	-31.7	95.0	-36.1	96.9
				-21.1	77.7	5.8	2.0	-25.2	79.2	-28.6	80.4	-31.9	81.5	-36.1	83.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.9	12.6	16.0	26.7	41.7	55.5	64.4	68.7	67.0	59.4	46.6	33.9	21.0	
	DBStd	21.53	13.10	11.40	10.52	8.40	7.53	6.53	4.94	5.33	7.30	7.77	8.99	11.31	
	HDD50	4691	1159	951	723	273	30	1	0	0	8	162	484	900	
	HDD65	8405	1624	1371	1187	701	308	89	19	41	197	571	932	1365	
	CDD50	2109	0	0	1	22	199	432	580	526	291	56	2	0	
	CDD65	352	0	0	0	1	13	71	134	102	30	1	0	0	
	CDH74	2643	0	0	1	12	158	582	977	693	215	5	0	0	
	CDH80	575	0	0	0	1	30	136	223	140	45	0	0	0	
Wind	WSAvg	6.1	6.9	6.9	7.0	7.3	6.5	5.4	4.9	4.6	4.8	5.8	6.2	6.4	
Precipitation	PrecAvg	43.1	3.6	2.6	3.0	3.6	3.7	4.0	4.1	3.4	3.6	4.1	3.5	3.5	
	PrecMax	57.3	5.8	4.1	5.2	6.3	7.5	6.2	7.2	5.1	6.0	7.4	5.4	6.5	
	PrecMin	33.7	1.2	1.3	1.3	0.9	2.0	1.7	1.6	1.2	1.1	1.3	1.2	1.5	
	PrecStd	5.8	1.3	0.7	1.0	1.7	1.4	1.3	1.3	1.1	1.2	1.9	1.0	1.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.9	43.5	57.0	74.5	83.6	88.0	89.4	87.8	85.8	73.7	60.1	47.4	
		MCWB	41.4	39.1	48.9	59.1	67.4	73.3	74.4	73.0	71.3	62.9	54.9	44.8	
	2%	DB	37.7	38.6	48.2	66.5	78.4	84.0	85.3	83.7	79.6	67.8	54.9	41.7	
		MCWB	35.7	35.5	41.3	52.5	63.0	70.3	72.2	70.6	68.6	59.8	50.9	39.6	
	5%	DB	34.7	35.6	43.1	60.5	74.0	80.4	82.2	80.7	75.6	62.5	50.1	37.7	
		MCWB	33.3	33.4	37.5	49.1	60.3	67.7	70.1	68.8	66.1	55.6	46.1	35.5	
	10%	DB	31.7	33.0	39.8	55.5	69.6	77.1	79.5	77.9	71.9	58.5	46.1	34.9	
		MCWB	30.4	31.1	35.2	45.7	57.9	65.3	68.4	67.5	64.1	53.1	42.6	33.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.5	40.0	51.3	61.5	70.2	75.9	77.4	75.2	73.3	65.8	56.4	45.6
			MCDB	42.9	42.2	55.1	72.1	80.3	85.2	86.5	84.5	81.6	71.2	58.5	46.8
2%		WB	36.1	36.0	42.4	54.8	66.2	72.4	74.3	72.7	70.2	60.5	51.5	40.0	
		MCDB	36.9	37.9	46.9	62.2	74.4	81.2	82.1	80.4	77.0	65.7	54.4	41.4	
5%		WB	33.3	33.6	38.4	50.5	63.0	69.8	72.2	70.8	67.8	57.0	46.5	36.0	
		MCDB	34.6	35.3	42.2	58.3	70.0	76.9	79.4	77.8	74.1	61.7	49.2	37.4	
10%		WB	30.1	31.2	35.8	47.0	59.8	67.5	70.2	69.2	65.4	53.8	43.1	33.3	
		MCDB	31.7	33.2	39.0	54.2	67.3	74.5	77.1	75.7	70.8	57.4	45.7	34.5	
Mean Daily Temperature Range	5% DB	MDBR	15.4	16.4	15.4	17.6	19.7	18.9	18.5	18.9	19.1	16.0	13.2	12.9	
		MCDBR	16.2	14.3	18.2	27.5	26.5	23.6	21.6	21.4	22.6	22.5	19.2	13.0	
		MCWBR	15.8	13.3	13.2	17.1	14.7	12.8	11.8	11.3	13.2	15.0	16.1	12.1	
	5% WB	MCDBR	15.9	14.1	16.8	23.9	21.6	19.9	19.3	18.8	20.5	20.3	18.3	12.9	
		MCWBR	15.7	13.6	13.2	16.5	14.0	12.3	11.6	11.0	12.8	14.8	16.2	12.4	
Clear-Sky Solar Irradiance	taub	0.255	0.285	0.329	0.354	0.382	0.435	0.420	0.407	0.368	0.335	0.303	0.283		
	taud	2.348	2.253	2.264	2.378	2.361	2.216	2.265	2.299	2.399	2.497	2.495	2.329		
	Ebn at Noon	274	283	285	287	281	265	268	267	269	264	254	246		
	Edh at Noon	23	32	36	36	38	44	41	38	32	25	21	22		
All-Sky Solar Radiation	RadAvg	448	753	1163	1413	1632	1786	1798	1588	1246	726	439	317		
	RadStd	55	78	100	148	158	134	145	75	103	64	54	25		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (43 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page